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SYSTEMATIC REVIEW

Effects of Combined Uphill–Downhill Sprinting Versus Resisted Sprinting Methods on Sprint Performance: A Systematic Review and Meta-analysis

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Abstract

Background Two specific sprint training methods that are present to varying degrees in research and practice are combined uphill–downhill sprinting (UDS) and resisted sprint training methods (RS). Both methods seem to improve sprint performance, but to the author's knowledge a comparison does not exist investigating the differences between the two training protocols and traditional sprinting.

Objective The present systematic review and meta-analysis investigated sprint performance changes between combined uphill–downhill sprinting and resisted sprinting methods (sleds, cables/bands, vests, uphill) and how these compared with traditional sprinting.

Methods A literature search was performed on 19 December 2022, in the databases PubMed, SPORTDiscus, Web of Science and SCOPUS, which from 22 studies yielded a total of 24 eligible groups (UDS, $n=6$; RS, $n=18$). Studies that measured sprint performance, had a traditional sprinting control, and used either training intervention in healthy individuals of any age for ≥ 4 weeks were eligible for the meta-analysis. The change in sprint performance from baseline to post intervention was compared between the interventions and their traditional sprinting control group. Outcomes were expressed as standardized mean differences (SMD).

Results The standardized changes in sprint performance between intervention groups and traditional-sprinting controls (negative in favour of intervention, positive in favour of traditional sprint) and 95% confidence interval (CI) were as follows: small for UDS (SMD −0.41 [−0.79, −0.03]; $p=0.03$), trivial for RS (SMD −0.14 [−0.36, 0.07]; $p=0.19$).

Conclusion Combined uphill–downhill sprinting was more effective than traditional sprinting, while resisted sprinting was not. It appears that resisted sprint interventions do not increase sprint performance any more than traditional sprinting. Subgroup analysis and meta-regression appear to show differences between sled loads and possible differences across distances tested. The results of this review and meta-analysis seem to warrant further investigations into the possibility that UDS may be a superior sprint training method to resisted and traditional sprinting.

1 Introduction

A variety of physical factors contribute to the success of sport participants, including change of direction, sprint performance, aerobic capacity, and strength and power [1–5]. Of these, sprint performance can be considered one of the most important physical qualities. In team sports such as football (soccer), nearly 50% of goals at the elite level are preceded by a straight sprint [6]. Additionally, sprint performance of professional athletes is continually improving [7], while the quantity and intensity of sprint actions continually increase [8]. Therefore, it is understandable that sprint training must also continue to match the evolving demands seen in sport settings.

Various sprint training programmes and methods exist in literature and practice which can vary in their modalities. Previous work by Rumpf et al. [9] and Plisk [10] classified sprint training methods on the basis of their specificity to sprinting. Sprint training can be categorized as either primary sprint training (traditional sprinting and technical drills without the use of external loads) or secondary sprint training (sprinting which is modified with assistance or resistance) and tertiary methods (non-specific methods such as strength and power training) [10]. Common secondary sprinting methods including (i) resisted sprinting using towing devices (sleds, cables), uphill sprinting or

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Key Points

Both resisted sprinting and combined uphill–downhill sprinting improve sprint performance.

Resisted sprinting was not significantly more effective than the control (traditional sprinting).

Combined uphill–downhill sprinting was significantly more effective than traditional sprinting (small differences).

Combined uphill–downhill sprinting appears to be a more effective sprint training method. As the characteristics of a hill cannot be altered, combined uphill–downhill sprinting is limited by the inability to modify the overload and overspeed properties like that seen with other methods (cables, sleds, vests).

vests and (ii) assisted and overspeed sprinting using towing devices, or downhill sprinting and both are used to varying degrees within research and practice [9, 11–14].

Resisted sprinting (RS) is defined as sprinting with the addition of an applied resistance in the forms of bands (with or without partner resistance), motorized cables (1080 Sprint), uphill sprinting, sleds or weights which apply overload to the athlete to achieve desired adaptations [9, 11, 15]. Assisted and overspeed strategies involve some form of assistance to the athlete using pulleys, bands, motorized cables (1080 Sprint) or downhill running, which reduce the load and conversely increase the velocity of movement [12–14]. While longitudinal studies are limited, the literature on assisted sprinting has studied the acute effects of training, showing acute enhancements in ground contact time, velocity and stride length when sprinting on flat ground [16–18]. Alternative methods can utilize both assisted and resisted forms of sprinting to enhance both load and velocity within a single repetition, and such an example is seen in combined uphill–downhill sprinting [19–23]. This alternative sprint-training method can be referred to as combined uphill–downhill sprinting (UDS), and involves sprinting both up and down artificial or natural hills or slopes within a single repetition, to impose an increase in load and velocity, and is a training strategy within the literature and practice to enhance sprint performance [19, 20, 22, 24–26].

While resisted strategies have shown improvements in sprint performance, it seems there is not a significant difference between resisted sprinting (secondary) and traditional sprinting (primary), as both show similar improvements in performance [11, 27, 28]. Nonetheless, logistical differences in implementation and requirements between resisted and unresisted sprint training modes should be considered [11, 29]. Regarding resisted sprinting variations, since many inherently require physical devices such as bands, weights and sleds, this can be a logistical and or financial barrier-to-entry for some sports teams and practitioners with limited space, budgets and equipment. Thus, this may potentially limit implementation and adherence of these programmes within large-team environments, and therefore an equally effective method that is more feasible may provide benefit to coaches. For these reasons, an alternative secondary sprint training method is of interest for investigation and comparison.

Regarding combined uphill–downhill sprinting, the literature is still unclear on how this training method compares with other specific sprint training programmes such as resisted sprinting. To the authors' knowledge, a comparison between combined uphill–downhill sprinting and resisted sprinting has not been done to evaluate differences in sprint adaptations. While several studies contain a traditional sprinting group, not all sprint intervention studies contain one as their control. For this reason, a systematic investigation is warranted that utilizes studies that contain a traditional sprinting control group. Therefore, the aim of this article is to review the relevant body of literature and to systematically analyse data from appropriate combined uphill–downhill sprinting and resisted sprinting studies in healthy individuals, to determine any differences and identify which method is more effective.

2 Methods

2.1 Information Sources and Search Strategy

The protocol of this review and meta-analysis was registered with the Open Science Framework on 14 December 2022 (https://osf.io/y8xzs/?view_only=da25219db91b452eb612e66a67fb4d80). This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30]. A search of the literature was done on 19 December 2022, of the following databases: PubMed, SPORTDiscus, Web of Science and Scopus. The following search string was input and adapted to each database: ((uphill OR up hill OR downhill OR down hill OR slope OR gradient) AND sprint*) OR ((Resisted OR loaded OR sled OR banded OR weighted) AND sprint*) NOT (rehab* OR Injur*). Following retrieval and filtering of the search, reference lists from applicable articles were read to find other relevant articles for review.

2.2 Selection Process and Eligibility Criteria

All collection, identification and screening was done by one researcher (M.J.H.) according to the inclusion and exclusion criteria. Reviewers (E.S.V. and P.A.) were consulted for article(s) that required further assessment for review. Endnote reference manager was used as the automation tool for collection, and management of duplicates.

The criteria for eligibility of articles are available in Table 1.

2.3 Data Item Extraction

One author, M.J.H., extracted data from the included studies, and E.S.V. reviewed and verified extracted data. The following data were extracted: sample level (non-sporting, recreational, amateur, sub-elite, elite etc.), sport (if applicable), age, sex, intervention type, load if RS, frequency and duration, (session *x* weeks), session volume (m), weekly volume (m), total volume (m), degree of slope, UDS platform dimensions, sprint assessment instrument, test-distance and outcome (*t*, time; V_a, average velocity; MRS, maximum running speed). Additional quantitative data were extracted to calculate SMD between groups and sessional training efficacy (TE_S). Sessional training efficacy is defined as the standardized change in sprint performance per training session and is calculated by dividing the SMD for each study's intervention group by the study's total number of training session [9, 31]. For clarity, levels were defined as follows: non-competitive (C), if they did not engage in regular sport practice or competition; amateur (B), if they practiced and competed their sport at an amateur club, university or college or if they competed but elite or professional status was not mentioned in the study; elite (A), if subjects practiced and competed professionally or semi-professionally.

2.4 Study Quality and Risk-of-Bias Assessment

To assess the risk of bias and methodological quality of the included studies, the revised Cochrane Risk of Bias tool for randomized trials was used to assess study bias and quality [32]. M.J.H. assessed each study and provided a score which was then reviewed and independently assessed by E.S.V. and P.A. The tool assesses study quality through assessments of five domains of possible bias: (1) randomization process; (2) deviations from intended interventions; (3) missing outcome data; (4) measurement of the outcome; (5) selection of the reported result. After review, each section can receive a score of 'low', 'some concern' or 'high', based on its study characteristics. Following assessment of each domain, an overall assessment is done, by assigning a score equivalent to the highest score set in any of the domains. A risk-of-bias assessment can provide information about study quality (presence of methodological safeguards) as well as bias (implications of safeguards on results) [33]. Assessment and

Table 1 Eligibility criteria for systematic review

Inclusion criteria
Randomized peer-reviewed studies with full-text availability
English language, without date restrictions
Healthy participants of any age or sex
Combined uphill-downhill sprinting (sprinting up and down hill within a single repetition, on any surface with a slope ≥ 1°) or resisted sprinting program (sleds, bands, cables, weighted vests, uphill) involving maximum sprinting
At least once per week, for ≥ 4 weeks, without a minimum sprinting distance requirement
Measured sprint performance as peak or average velocity or sprint time Measured using an automated device, such as a photocells, radar, or video analysis
Study included a control group that performed traditional sprinting (flat, unresisted)
Exclusion criteria
Meta-analyses, systematic reviews, narrative reviews
Unavailable text, not peer-reviewed or not in English
Studies with unhealthy participants (injury, pathology.)
Without sloped sprinting or resisted sprinting group Testing protocol and training program not described
Acute study (< 4 weeks)
Intervention group performed repeat sprints, instead of single-effort sprints without adequate rest between repetitions
Sprint performance was not measured, or done using manual devices (stopwatch) Data unavailable
Sprint intervention group performed other exercises (plyometrics, change of direction, strength-training)
No control group that performed traditional sprinting was present

visualizations (traffic light plot and summary were done using Risk-of-bias VISualization (robvis) [34]. Details of the assessment of study quality can be found in the electronic supplementary material (ESM), S1 and S2.

2.5 Data Synthesis and Effect Measures

The outcome included in this review was the change in sprint performance from baseline to post intervention between groups. Effect sizes are included in each table (Tables 2, 3), as well as qualitative assessments of magnitudes. Since some studies measured sprint time and others measured velocity, sprint velocity outcomes were multiplied by − 1 so that improvements for both metrics were in the same direction. Standardized mean differences (SMD) between groups were calculated as the intervention group minus the control group, and thus negative values signify the direction of the intervention (RS or UDS), and positive values in favour of the control (traditional sprint). Reference values were then based upon absolute effect size, and are as follows, < 0.2, trivial; 0.2–0.6, small; 0.6–1.2, moderate; 1.2–2.0, large; 2.0–4.0 very large; > 4.0 nearly perfect [35]. SMD was calculated using adjusted Hedges' *g* (which uses a correction factor to adjust for small sample bias). The equations for the above outcomes as well as TE_S are described below.

$$\mathrm{SMD}_{\text{within group}} = \frac{\text{Mean Post} - \text{Mean Pre}}{\text{Pooled SD}} \times \text{ small sample correction factor}$$

$$\mathrm{SMD}_{\text{between groups}} = \frac{\text{Intervention group ES} - \text{Control Group ES}}{\text{Pooled SD}} \times \text{ small sample correction factor}$$

$$\text{Pooled SD} = \sqrt{\frac{(n_1 - 1)\mathrm{SD}_1^2 + (n_2 - 1)\mathrm{SD}_2^2)}{(n_1 + n_2 - 2)}}$$

$$\text{Small sample correction factor} = 1 - \frac{3}{4(n_1 + n_2) - 9}$$

$$\mathrm{TE}_S = \frac{\mathrm{SMD}_{\text{between groups}}}{n_{\text{trainingsessions}}}.$$

2.6 Meta-analysis

A random-effects meta-analysis was conducted to compare the differences between interventions. A random-effects meta-analysis assumes some differences in effects exist due to chance as well as variation in the true intervention effect [36]. Studies containing a control group that performed traditional (unresisted and horizontal) sprinting were analysed so improvements could be compared between intervention groups (combined uphill–downhill or resisted) and control groups (traditional sprinting). The meta-analysis was conducted using the reference manager software, RevMan [37]. A sensitivity analysis was done by completing additional meta-analyses using a fixed-effect model. Pre and post means and standard deviations were used to determine within-group SMD. These effects were then compared between groups (intervention and control groups) to determine the SMD between groups.

2.7 Certainty Assessment

An assessment of certainty of the data of the meta-analysis was done using the GRADE approach using the GRADEpro software [38]. The GRADE approach assesses certainty of findings in five domains (risk of bias, inconsistency of results, indirectness of evidence, imprecision, publication bias). Assessments were independently done by M.J.H. and E.S.V., any differences in assessments were reviewed and the lower certainty score was assigned. Certainty assessments was done for the question of the meta-analyses, 'Intervention compared to traditional sprinting for sprint performance'. A summary of findings was then generated using the software [38] and is included in the ESM (S5).

3 Results

3.1 Search Result

After conducting the search of the databases, a total of 4559 records were identified. After removal of duplicates, 3322 records remained to be screened on the basis of title and abstract and the inclusion and exclusion criteria previously described. After title and abstract screening, 48 records remained which were then sought for retrieval. Forty-two full-text articles were retrieved and thus assessed for eligibility. From the assessed reports, 21 studies remained. One study was then added from other sources, assessed and included, resulting in 22 studies

Table 2 Characteristics of combined uphill–downhill sprinting (UDS) interventions

Study	Sample characteristics				Training program						Testing		Training effect		
	n (sex: M/F)	Age	Level	Sport	Sl	Freq × weeks	Session dist (m)	Week dist (m)	Total dist (m)	Platform dimensions (m) (↔/↑/↔/↓/↔)	Instrument (outcome)	Dist (m)	SMD between groups (− in favour of intervention + in favour of control)	Δ	TE_S ($\frac{SMD}{n_{sessions}}$)
Bissas et al. [21]	14 (Mixed)	23.9 ± 3.3	C	NA	3	3 × 6	480–540	1440–1620	9000	10/20/10/20/20	V/C (MRS)	35	− 0.35	Sm	− 0.02
Cetin et al. [23]	6 (M)	20.8 ± 1.5	C	NA	4	3 × 6	680–850	1200–2550	6885	15/20/15/20/5	Ph (*t*)	100	− 0.41	Sm	0.02
Paradisis and Cooke [25]	7 (Mixed)	23.8 ± 1.4	C	NA	3	3 × 6	480–640	1440	9600	10/20/10/20/20	V/C (MRS)	35	− 0.29	Sm	− 0.02
Paradisis et al. [20]	18 (M)	24.1 ± 2.1	C	NA	3	3 × 8	480–800	1440–2400	15,360	10/20/10/20/20	V/C (MRS)	35	− 0.36	Sm	− 0.02
Paradisis et al. [19]	8 (mixed)	24.5 ± 2.0	C	NA	3	3 × 8	480–800	1400–2400	15,360	10/20/10/20/20	V/C (MRS)	35	− 0.70	Mod	− 0.03
Paradisis et al. [22]	6 (mixed)	25.3 ± 2.9	B	Sprint	3	3 × 6	480	1440–2400	8640	10/20/10/20/20	V/C (MRS)	35	− 0.42	Sm	− 0.02

SMD reference: < 0.2, trivial; 0.2–0.6, small; 0.6–1.2, moderate; 1.2–2.0, large; 2.0–4.0, (negative values in favour of intervention)

Δ: qualitative change; (↔/↑/↔/↓/↔): the sequential horizontal, uphill, horizontal, downhill, horizontal sections of the sprint platform, *A* pro or semi pro, *B* amateur, *C* non practicing/competing, *dist* distance, *F* female, *Freq* frequency of training (number of sessions per week), *M* male, *MRS* maximum running speed, *NA* not applicable, *Ph* Photocell, *Sl* slope degree, *SMD* standardized mean difference, *t* time, *V/C* video/computation, *Sm* small, *Mod* moderate

Table 3 Characteristics of resisted sprinting (RS) interventions

Study	Sample characteristics					Training program					Testing		Training effect		
	n (sex: M/F)	Age	Level	Sport	Type	Load	Freq × weeks	Session dist (m)	Week dist (m)	Total dist (m)	Instrument (outcome)	Dist (m)	SMD between groups (− in favour of intervention + in favour of control)	Δ	TE_S ($\frac{SMD}{n_{sessions}}$)
Alcaraz et al. [51]	11 (mixed)	21.5 ± 2.2	A	Sprint	S	7.5% VL	2×4	90–180	540–980	1080	Radar (V_a)	15	+0.05	Triv	0.01
												30	− 0.08	Triv	− 0.01
												50	− 0.15	Triv	− 0.02
Chaalali et al. [40]	7 (M)	16.6 ± 0.2	A	Soccer	RCP	NA	2×6	90–360	90–360	2520	Ph (*t*)	5	− 0.65	Mod	− 0.05
												10	− 0.75	Mod	− 0.06
												20	− 0.20	Sm	− 0.02
												30	− 0.15	Triv	0.01
Clark et al. [41]	7 (M)	19.73 ± 1.01	B	Lacrosse	S	10.2% BM	2×7	220–402	475–805	4060	Ph (*t*)	36.6	+0.42	Sm	− 0.03
	6	19.79 ± 0.92			WV	18.5% BM							+0.21	Sm	0.01
Derakhti et al. [52]	8 (M)	15.7 ± 0.5	A	Soccer	RC	50% VL	2×4	100	200	800	1080 Sprint	5	− 0.90	Mod	− 0.11
											(*t*)	10	− 0.86	Mod	− 0.11
												20	− 0.82	Mod	− 0.10
												30	− 0.59	Sm	0.07
Lockie et al. [42]	9 (M)	23.1 ± 4.2	B	Field Sports	S	12.6% BM	2×6	195–320	390–640	3100	Veloc (V_a)	5	− 0.04	Triv	0.00
												10	− 0.17	Triv	− 0.01
Luteberget et al. [43]	9 (F)	20.4 ± 3.1	A	Handball	S	12.4% BM	2×10	240–280	480–560	4680	Ph (*t*)	10	+0.58	Sm	0.03
												30	+0.60	Mod	− 0.03
McMorrow et al. [39]	6 (M)	25.5 ± 3.2	A	Soccer	S	30% BM	2×6	100–180	200–360	1360	Ph (*t*)	5	− 0.16	Triv	− 0.01
												10	− 0.30	Sm	− 0.03
												20	− 0.25	Sm	0.03
Morin et al. [44]	10 (M)	26 ± 4.0	B	Soccer	S	80% BM	2×8	200	400	3200	Radar (*t*)	5	− 0.36	Sm	− 0.02
												20	− 0.15	Triv	0.01
Paradisis and Cooke [25]	7 (Mixed)	23.8 ± 1.4	C	NA	Uphill	3°	3×6	480–560	1400–1680	9120	V/C (MRS)	35	0.11	Triv	0.00

Table 3 (continued)

Study	Sample characteristics					Training program					Testing		Training effect		
	n (sex: M/F)	Age	Level	Sport	Type	Load	Freq × weeks	Session dist (m)	Week dist (m)	Total dist (m)	Instrument (outcome)	Dist (m)	SMD between groups (− in favour of intervention + in favour of control)	Δ	TE_S ($\frac{SMD}{n_{sessions}}$)
Pareja-Blanco et al. [45]	14 (F)	22.0 ± 1.5	C	NA	S	40% BM	1 × 8	120–240	120–240	1440	Ph (*t*)	10	0.00	NA	0.00
												20	0.00	NA	0.00
												30	− 0.12	Triv	0.02
Rey et al. [46]	10 (M)	23.6 ± 2.7	B	Soccer	WV	18.9% BM	2 × 6	120–280	240–560	2560	Ph (*t*)	10	+ 0.45	Sm	0.04
												30	− 0.26	Sm	0.02
Rodrıguez-Rosell et al. [29]	12 (M)	23.8 ± 2.8	B	Mixed Sports	S	20% BM (14% VL)	1 × 8	120–240	120–240	1440	Ph (*t*)	10	+ 0.17	Triv	0.02
												20	+ 0.36	Sm	0.05
												30	+ 0.24	Sm	0.03
	12	22.1 ± 1.7				40% BM (24% VL)						10	− 0.35	Sm	− 0.04
												20	− 0.10	Triv	− 0.01
												30	− 0.20	Sm	− 0.03
	12	21.9 ± 2.2				60% BM (33% VL)						10	− 0.15	Triv	− 0.02
												20	0.00	NA	0.00
												30	− 0.06	Triv	− 0.01
	12	22.2 ± 1.8				80% BM (39% VL)						10	0.00	NA	0.00
												20	+ 0.15	Triv	0.02
												30	+ 0.05	Triv	− 0.01
Sinclair et al. [47]	13 (M)	18.9 ± 0.5	B	Rugby	S	20% VL	2 × 8	120–180	240–360	1920	Ph (*t*)	5	− 0.43	Sm	− 0.03
												10	− 0.32	Sm	− 0.02
												20	− 0.25	Sm	0.02
Spinks et al. [48]	10 (M)	21.8 ± 4.2	B	Football Code	S	10% VL	2 × 8	215–340	430–680	4090	Veloc (V_a)	5	− 0.16	Triv	− 0.01
												15	− 0.29	Sm	− 0.02
Upton [13]	9 (F)	19.6 ± 0.9	B	Soccer	RC	12.6% BM	3 × 4	183	549	1644	Ph (V_a)	4.6	− 0.08	Triv	− 0.01
												13.7	+ 0.19	Triv	0.02

Table 3 (continued)

Study	Sample characteristics					Training program					Testing		Training effect		
	n (sex: M/F)	Age	Level	Sport	Type	Load	Freq × weeks	Session dist (m)	Week dist (m)	Total dist (m)	Instrument (outcome)	Dist (m)	SMD between groups (− in favour of intervention + in favour of control)	Δ	TE_S ($\frac{SMD}{n_{sessions}}$)
												22.9	− 0.04	Triv	0.00
												36.6	− 0.16	Triv	− 0.01
West et al. [49]	10 (M)	26.8 ± 3.0	A	Rugby	S	12.6% BM	2 × 6	240	480	2880	Ph (*t*)	10	− 0.23	Sm	− 0.02
												30	− 0.28	Sm	0.02
Zafeiridis et al. [50]	11 (M)	20.1 ± 1.9	B	Team Sports	S	5 kg	3 × 8	280	840	6720	Ph (V_a)	10	− 1.30	Lrg	− 0.05
												20	− 0.79	Mod	− 0.03

(Negative values in favour of intervention) SMD reference: < 0.2, trivial; 0.2–0.6, small; 0.6–1.2, moderate; 1.2–2.0, large; 2.0–4.0

Δ: qualitative change, *A* pro or semi pro, *B* amateur, *BM* body mass, *C* non-practicing/ competing, *Dist* distance, *Freq* frequency of training (number of sessions per week), *F* female, *M* male, *MRS* maximum running speed, *NA* not applicable, *Ph* photocells, *RC* resistance cable, *RCP* resistance cable + partner resistance, *S* sled, *SMD* standardized mean difference, *t* time, TE_S sessional training efficacy, *U* Uphill, *V/C* video analysis and computation, V_a average velocity, V_m max velocity, *Veloc* velocimeter, *Vol* volume, *VL* velocity loss, *WV* weighted vest, *Triv* trivial, *Sm* small, *Mod* moderate, *Lrg* large

used in the systematic review (Fig. 1) One study [39] met the inclusion criteria for this review but did not include raw data, and therefore had to be extracted using graph digitizing software (PlotDigitizer, 3.1.5, 2022, [2–5]).

3.2 General Study Characteristics

The literature search yielded a total of 22 studies from which 24 groups were used in the analyses ($n = 6$ for UDS [19–23, 25] and $n = 18$ for RS [13, 29, 39–52]). Each study's characteristics regarding participants, training programmes, and testing and outcome are listed in Tables 2 and 3. The most common RS method was sled sprinting. Resistive loads varied, ranging from % velocity loss (VL) (7.5–50%), % body mass (BM) (10.2–80%) or absolute loads (5 kg). Sprint testing was assessed in the studies through various automated methods such as, photocells, radar guns [42, 44, 48, 51], video analysis and computation [19–22] or the use of the 1080 Sprint [52]. Within UDS groups, the degree of each slope was also defined in Table 2, with values ranging from 3° to 4°. Two different platform types were used among the UDS studies. The segment distances for each platform are in Table 2. Additionally, Fig. 2 contains the combined uphill–downhill platforms used in the respective studies. Performance metrics across studies were either average velocity, peak velocity achieved during testing distance, time to complete sprint test or maximum running speed (used in several studies from the same author group) [19–22]. The levels of subjects were more varied in RS than UDS, as five UDS studies contained samples that were non-practising or competing and one UDS contained amateur athletes. RS studies contained all three levels (two non-practising/competing, nine amateur; and six elite). Test distances between studies ranged from 4.6 to 100 m, and the most frequently used distance for sprint testing was 35 m and 30 m, for UDS and RS, respectively. Within each table, split-time outcomes (0–5, 0–10 etc.) are also listed whenever applicable. Regarding training programmes, total programme training volumes ranged from 800 to 9120 m for RS groups, and 6885 to 15,360 m for the UDS groups. Each study's training programme can be found in the ESM Fig. S7. TE_S is included for each study group and is found in Tables 2 and 3. The average TE_S for RS and UDS respectively was 0.002 and − 0.01 per session. Additionally included in Tables 2 and

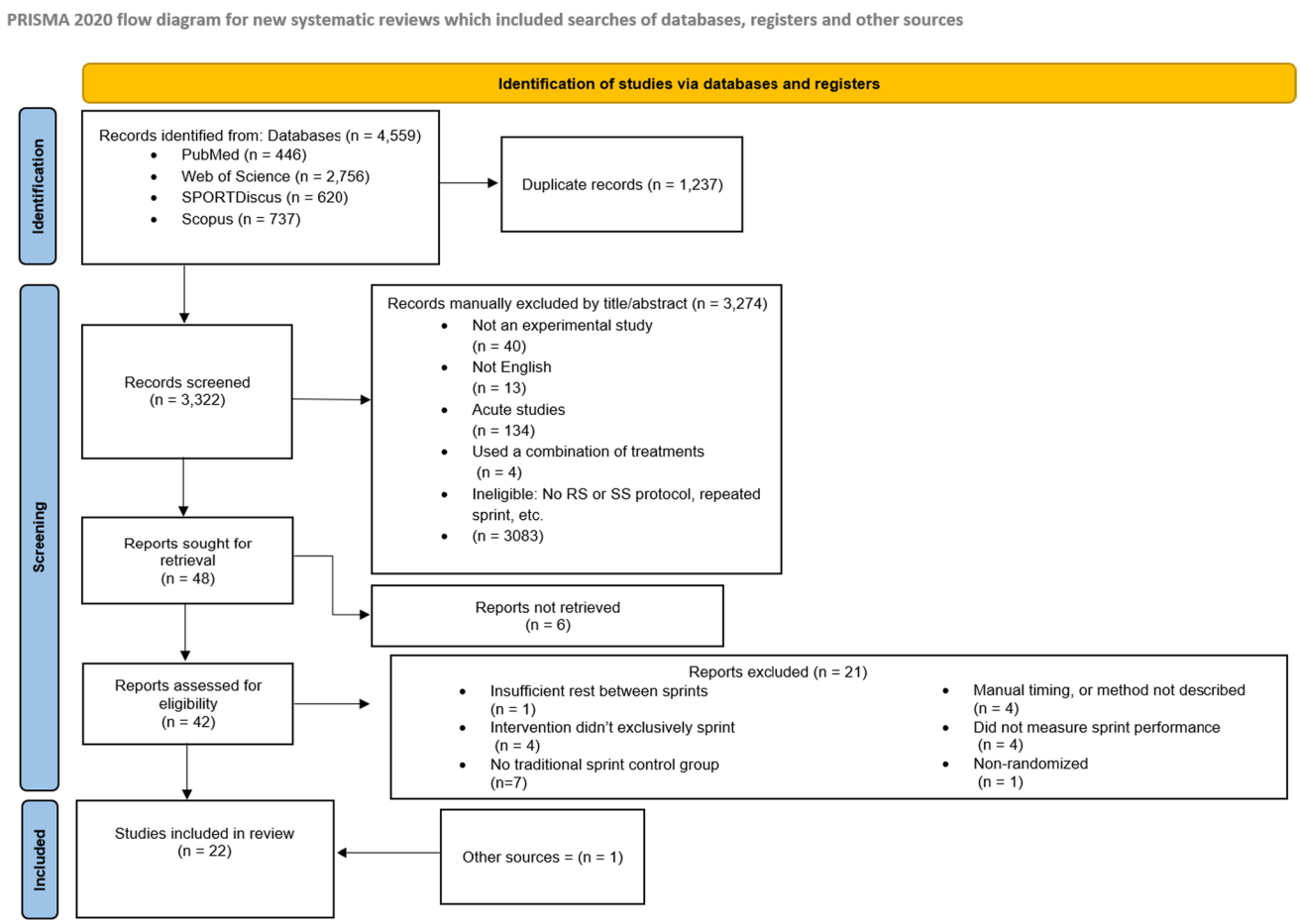

Fig. 1 The PRISMA flow diagram of selection process

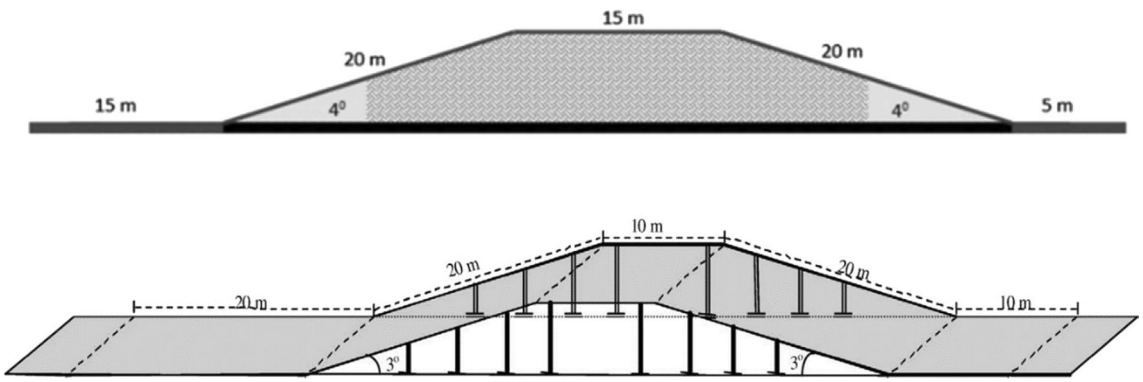

Fig. 2 The sprint platforms used in the combined uphill–downhill sprinting (UDS) interventions. Above: Cetin et al. [23]; below: Bissas et al. [21], Paradisis & Cooke [25], Paradisis et al. [20], Paradisis et al. [19] and Paradisis et al. [22]

3 are quantitative and qualitative descriptors of the training outcomes from each training intervention. SMD is presented as the between group effect size with negative values signifying in favour of the intervention group.

3.3 Study Quality and Risk-of-Bias Results

Results from the study quality and risk-of-bias assessment tools are shown in S1 and S2 of the ESM. Overall, risk of bias among studies was low. Risk of bias in all UDS studies was low, while in RS studies nearly 70% were low risk and roughly 30% had some concern. The most common domains with concern were bias due to the randomization process and missing outcome data from attrition.

3.4 Meta-analysis Results

A random-effects model forest plot of the SMD and 95% confidence intervals between interventions groups (UDS and RS) and control groups (traditional sprinting) is available in Fig. 3. Only studies that had a control group that performed traditional sprinting were included. Overall, heterogeneity was low in UDS and RS ($I^2 = 0$) [36]. The studies included in this review contain clinical variability in participants (age, sex, level), interventions (UDS or RS) and outcomes (sprint distance). Methodological homogeneity was preserved as best as possible with our review's eligibility criteria of study design, and the risk-of-bias assessments.

Funnel plots of the analysis are included in the ESM Fig. S3. Funnel plot did not show signs of publication bias (asymmetry or clustering away from the mean) for RS studies. UDS studies showed some slight asymmetry from the mean, although a larger sample of studies is required to more conclusively detect publication bias in this group.

3.4.1 Interventions Compared with Traditional Sprinting

The combined uphill–downhill studies contained a total of six intervention groups (Fig. 3). One combined uphill–downhill study contained an uphill-only group [25] and was included in the resisted sprinting group; thus, RS had 18 intervention groups (Fig. 3). If studies contained multiple loads (for example, 10% and 20% BM), the highest load was kept. Studies that lacked a group that performed traditional sprinting were excluded from the comparison. Changes in sprint performance were compared between intervention and traditional sprint groups, for the longest measured sprint distance. If a study contained multiple sprint distance measurements (5, 10, 20 m), the longest distance was used in the analysis, as multiple distance sprint outcomes of the same group are highly correlated [53].

Combined uphill–downhill sprinting was significantly more effective than traditional sprinting (SMD − 0.41 [− 0.79, − 0.03]; $p = 0.03$; $I^2 = 0\%$), resisted sprinting was not more effective than traditional sprinting (SMD − 0.14 [− 0.36, 0.07]; $p = 0.19$; $I^2 = 0\%$). UDS effect sizes ranged from trivial to moderate, while effect sizes between RS and traditional sprinting ranged from small for RS to trivial for the control group. A forest plot of the result is shown in Fig. 3.

3.5 Sensitivity Analysis

A sensitivity analyses was done comparing the random-effects models previously conducted above with a fixed-effect model [54]. The additional model is in the ESM (Fig. S4). Sensitivity analyses revealed similar results when the fixed-effect model as considered in comparison with the random-effects models.

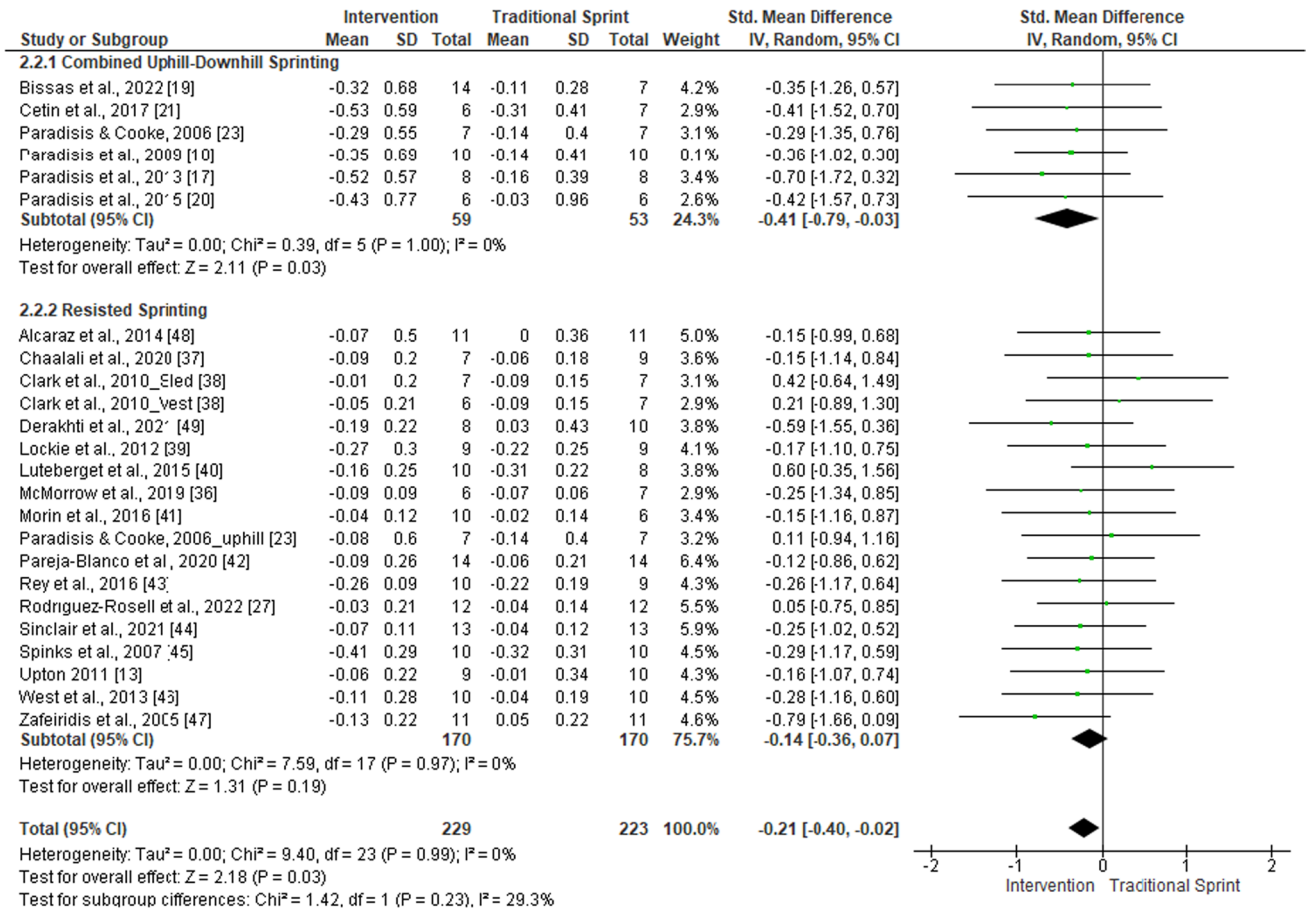

Fig. 3 Random-effects analysis of SMD performance change between intervention groups and traditional sprint controls. Green dots denote the estimates for each study; black diamonds denote the pooled effect among groups

Due to the small number of studies within UDS (6), further subgroup analyses were not done to avoid the risk of false positives. Additional comparisons increase the probability of finding significant *p* values by chance. For the objective of this review and meta-analysis, a broader analysis between groups was deemed acceptable.

3.6 Certainty of Evidence

The results of the GRADE assessments, including the level of certainty and reasoning, are in the summary of findings table in the ESM (S5). The level of certainty was high for both analyses.

3.7 Meta-regression

A meta-regression was done to further examine the moderating effects within analysed resisted sled sprinting studies. Sled sprinting data was compared across test distance and loading conditions to examine their moderating impacts on group effect size. This subgroup analysis was only done on sled sprint groups (as it was the most common resisted modality) to ensure consistency in load classification and since UDS group data were insufficient [36] and lacked heterogeneity in loading (slope). Thirteen studies [13, 29, 39, 41–45, 47–49, 51] comprising a total of 41 groups were included in the analysis. Weighted-vest studies were not included, and the 10 m effect size from Luterberget [43] was identified as an outlier using the rstatix package in R.Meta. Regression plots were done using the smplot package [55].

To perform the subgroup analysis and examine the load–velocity relationship across distances, sled data were examined for four different loading conditions (low, moderate, heavy, very heavy). The sled sprint groups from the included studies prescribed loading as either absolute load (kg), relative to body mass (% BM) or based on velocity loss (% VL) (Table 3). Classifications of loading were based on the system used by Zabaloy et al. [56] and are as follows:

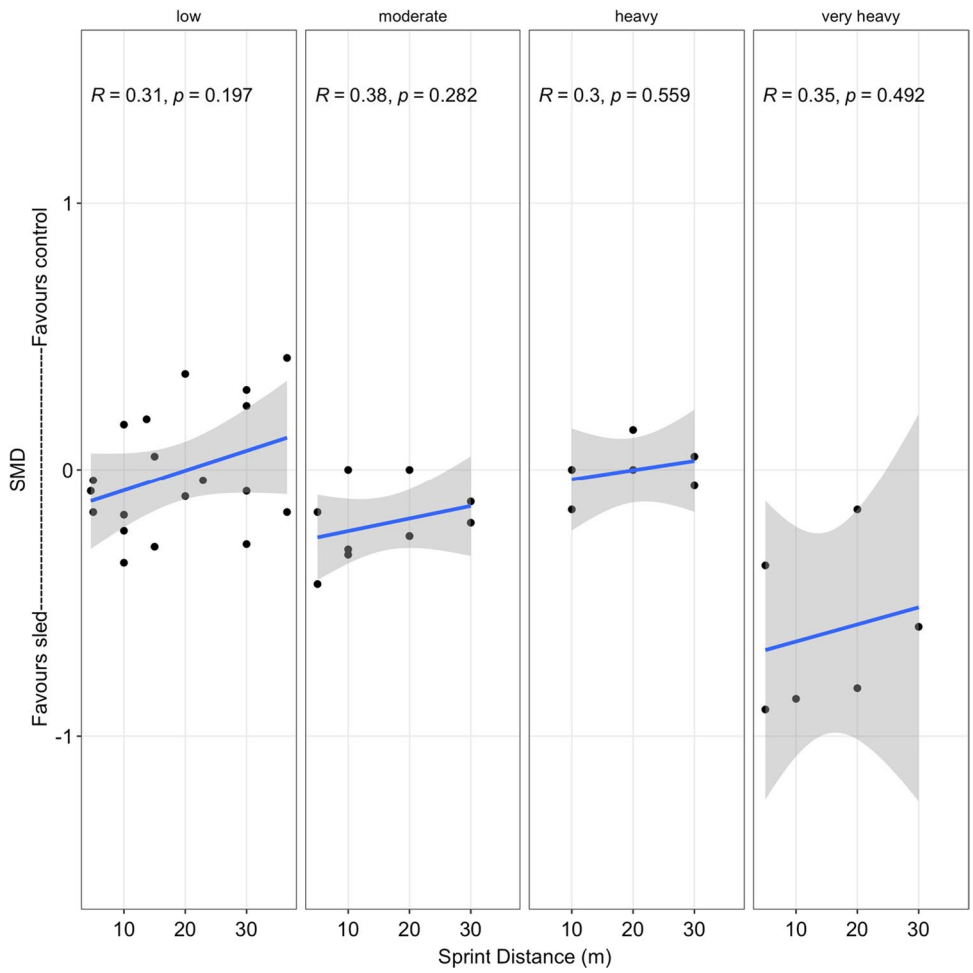

Fig. 4 Comparison of the SMD (negative in favour of RS, positive in favour of control group) and the sprint distances tested across each loading prescription

Table 4 Loading group descriptives

Load	Characteristic	Studies	Groups	Mean
Low	> 0–20% BM or 2.5–10% VL	[13, 16, 29, 42, 43, 48, 49, 51]	19	− 0.01
Moderate	> 20–50% BM or > 10 – < 30% VL	[29, 39, 45, 47]	10	− 0.20
Heavy	> 50–80% BM or 30 – < 50% VL	[29]	6	− 0.00
Very heavy	≥ 80% BM or ≥ 50% VL	[44, 52]	6	− 0.61

low (0–20% BM or 2.5–10% VL); moderate (> 21–50% BM or > 10 to < 30% VL), heavy (> 50–80% BM or 30 to < 50% VL), very heavy (≥ 80% BM or ≥ 50% VL). The results of the meta-regression are found in Fig. 4, Tables 4, 5 and the ESM S12 and S13.

From the results of the analysis there appears to be a weak positive relationship (Pearson, $r = 0.32$, $p = 0.043$) between sprint distance tested and SMD (not adjusted for load) such that the effect size of sled sprinting decreased as the tested sprint distance increased from 4.6 to 36.6 m (ESM S13). When adjusting the model for load, analysis

Table 5 ANCOVA results and post-hoc analysis of between-group differences for adjusted model

Term	Comparison	*p* (adjusted)
Distance * load	Low–moderate	0.22
	Low–heavy	1.00
	Low–very heavy	< 0.001
	Moderate–heavy	0.64
	Moderate–very heavy	0.002
	Heavy–very heavy	< 0.001

of covariance (ANCOVA) showed that the relationship between distance and SMD was not statistically significant ($p = 0.064$) (ESM S11). When adjusting the model for distance, ANCOVA results showed a statistically significant difference between groups ($p < 0.001$) (ESM S11). A post-hoc Tukey test (Tables 4, 5) revealed between-group differences in effect between the very heavy prescription and low, moderate and heavy loading conditions respectively (−0.61, $p < 0.001$; −0.50, $p = 0.002$; −0.61, $p < 0.001$). Differences between all other loading groups were not significant. Within each loading group (Fig. 4), the analysis also revealed weak positive linear relationships for low, moderate, heavy and very heavy groups respectively ($r = 0.31$, $p = 0.20$; $r = 0.38$, $p = 0.28$; $r = 0.3$, $p = 0.56$; $r = 0.35$, $p = 0.49$), these results were not statistically significant.

4 Discussion

The aim of the present review was to systematically analyse the available body of research to compare effects between combined uphill–downhill sprinting and resisted sprinting against traditional sprinting controls. Previous reviews have exclusively analysed sled sprinting [11, 27, 28], but a more extensive review of various resisted sprint methods and how these methods compare with combined assisted-resisted methods (uphill–downhill sprinting) was worthy of investigation. To the authors' knowledge, no review existed comparing sprint performance changes between combined uphill–downhill sprinting and resisted sprinting.

Presently, these findings seem to show that both combined uphill–downhill sprinting and resisted sprinting improve sprint performance. While further research is recommended to investigate combined uphill–downhill sprinting effects at shorter distances (< 35 m), our results correspond to previous findings [11]. The most recent review of resisted-sled sprinting [11] does seem to show that sled sprinting may be a more suitable training strategy for improving accelerative speed (< 20 m) compared with improving maximum velocity (> 20 m). Additionally, there seems to be evidence that UDS improvements are significantly greater than those of traditional sprinting (small differences for UDS, trivial differences for RS) (Fig. 2). This is interesting as both our analysis as well as the current literature [11, 27, 28] repeatedly show that resisted sprinting adaptations are trivial and non-significant compared with those of traditional sprinting.

A previous review and meta-analysis done in 2018 by Alcaraz and colleagues [11] showed that sled sprinting is an effective way to improve sprint performance especially in the early acceleration phase (< 20 m), while the effect tapered at distances ≥ 20 m, showing effects that were not statistically significant ([this is in line with the trend seen in our analysis (Tables 4, 5)]. When effects were compared with control groups that performed unresisted sprinting in that same review, the results revealed that the improvements were not significantly different between resisted and unresisted sprinting, independent of sprint phases [11]. Additionally, similar results were observed in another systematic review [27], when sleds and weighted-vest sprinting were compared with unresisted sprinting in young soccer players. It seems that, perhaps among resisted modalities, a variety of resisted sprinting programmes provide similar improvements in sprint performance [11, 27, 28]. The results of our review indeed show similar conclusions (Fig. 3). This is in line with the principles of training specificity and a previous review done by Rumpf and colleagues [9], showing that when specific training (resisted, unresisted, incline sprinting), non-specific training (resistance, plyometric training) and combined methods (specific and non-specific) were assessed, specific training showed the greatest benefit to sprint performance across distances 0–10, 0–20, 0–30 and 31 + m.

Studies involving general hill sprinting have shown success across differing levels of experience. In several studies, combined uphill–downhill training improved sprint performance by 4% in recreational adults and trained sprinters after 6–8 weeks [19, 20, 22]. Similar improvements were also seen in semi-professional field hockey players within only 4 weeks when only uphill sprint training was compared with a control group that followed their normal training programme, but did not sprint [26]. Thus, highlighting that general hill sprinting programmes may be effective across skill levels, sporting populations. Interestingly, in this review one study contained both a resisted (uphill) and combined uphill–downhill group [25]. The study saw significant improvements (pre to post intervention) in maximal running speed in only the combined uphill–downhill group, while there were no significant changes in the uphill and traditional sprinting groups. Additional kinematics were measured in the study and replicated in others, and it seems that performance adaptations are mainly manifested through an increased step rate [19, 20, 25].

It seems the mechanical and technical nature of resisted sprinting can additionally be used as a technical tool to develop sprint technique. With uphill sprinting, due to the inclination, the athlete's horizontal orientation with respect to the ground is prolonged causing an increase in horizontal (relative) ground reaction forces [57], allowing athletes to learn and develop the technical and kinetic demands of early acceleration [58]. The subgroup analysis and meta-regression in this review did not show significant differences between the 'low', 'moderate' or 'heavy' groups, while the 'very heavy' group demonstrated significant differences to all others (Fig. 4). The estimation is that perhaps the advantage that other resisted modalities such as sleds, vests and cables offer when compared with uphill sprinting

is the possibility of loading the sprint more heavily and thus emphasizing higher force production. This greater demand in force application as well as technical training may be appropriate goals within a periodized programme when targeting demands seen within early acceleration [44, 58], highlighting the utility of some heavier resisted sprinting methods.

At times throughout a periodized training programme, athletes may also be exposed to training methods that intentionally enhance movement velocity beyond that which is typically possible to achieve by the individual alone. Often referred to as 'overspeed' or 'assisted training', these methods have been studied in the literature and used within specific sprint training methods as well as non-specific modes (plyometrics) to enhance neuromuscular power [12, 24, 59–61]. Contrast training and, specifically, French contrast (a sequence involving a heavy load compound lift, plyometrics, light/moderate-load exercise and assisted plyometrics) involves the use of overspeed tactics to achieve these desired adaptations [60–62]. Assisted training, like that in French contrast, has shown positive adaptations both acutely in countermovement jump performance as well as in movement velocity after 8 weeks [59, 61]. While long-term studies utilizing overspeed tactics in sprint training are limited, acute effects from assisted sprinting with towing devices and hills indicate movement velocity is indeed enhanced during the sprint, mainly through an increase in stride length [12, 63, 64]. The individual studies in this review which utilized a combination of uphill and downhill sprinting (UDS) showed small to large improvements in sprint performance [20–22, 25, 57]. Only one study involved a group that trained exclusively downhill [25]; interestingly, this group experienced changes in sprint performance that were not statistically significant. It is important to note that, due to the advanced nature of assisted sprinting methods, strength and conditioning coaches should be strategic with the timing of their implementation within a programme. As assisted sprinting exposes athletes to supramaximal speeds, this method should be used with advanced-level athletes who have gone through a comprehensive preparatory phase. Exposing untrained or less-experienced athletes to supramaximal speeds may increase the likelihood of injury.

Results from the subgroup analysis and meta-regression for sled sprinting seem to demonstrate that, when not accounting for loading condition, there seems to be a weak positive relationship between the sprint distance tested and the effect size. These results seem to show that the improvements from sled sprinting decreased as for the tested distance increased (4.6–36.6 m). When not accounting for distance tested, small to moderate differences were seen between the very heavy loads ($\geq$ 80% BM or $\geq$ 50% VL) and all other groups, while differences between all other groups were not significant. Of note, the very heavy group also contained some reasonable variability due to the study by Derakhti et al. [52], which may have seen strong effects due to their sample being professional adolescent athletes. Additionally, when the relationship between test distance and SMD was analysed within each loading group (Fig. 4), the results of this meta-analysis and meta regression seem to indicate that the linear relationship further weakens within a loading prescription. The model which excludes load (ESM S13), seems to show that, for improving > 30 m performance, sled sprinting does not seem more effective than the traditional sprinting controls, which is similar to the findings from the literature [11]; yet considering the variability and weak relationship, it seems that the effects cannot be explained by distance alone. This is indeed further clarified when the model was adjusted for load (ESM S11), showing that the differences between distances were indeed not significant. Among the different load groups, there do appear to be differences between the very heavy load group and the low, moderate and heavy groups (Tables 4, 5), while the low, moderate and heavy groups did not show significant differences among each other. This finding was consistent in the adjusted and unadjusted models, but again due to the variability resulting from the recent Derakhti et al. study, this reinforces the need for further investigation to determine the superiority of one loading prescription over another.

One possible reason for the difference between combined uphill–downhill and resisted sprinting, (and that resisted sprinting seems more effective at distances < 20 m [11] could be related to the load–velocity stimuli of each respective training method. If it is considered that resisted sprinting overloads the sprint action thereby increasing demands in force application and ground contact times [65], this then may explain its greater effectiveness at improving the acceleration phase more than the maximum-velocity phase [11]. A benefit UDS may provide is that combined uphill–downhill sprint training programmes like those seen in the studies of this review may expose an individual to a broader load–velocity spectrum than loaded or resisted sprinting alone. It is possible that, since combined uphill–downhill sprinting includes both overload and overspeed components, the stimuli and relevant adaptations may correspond better to a full sprint which involves acceleration and maximum-velocity phases. Throughout sprinting, the force–velocity demands and manifestations share an inverse relationship. From a stationary start, sprint speed gradually increases before plateauing once maximum velocity is reached [66]. Conversely, the mean propulsive force produced by the individual is highest at the beginning (acceleration phase) when sufficient force must be generated to overcome inertia and accelerate the athlete forward, but then steadily decreases as factors such as ground contact times decrease [66]. Therefore, it can be theorized that since sprinting involves both an acceleration

phase involving high force relative to velocity demands [66, 67], and a maximum-velocity phase, which requires greater relative velocity demands, perhaps a sprint training programme that imposes both overload (force) and overspeed may be a more relevant training stimulus. It is hypothesized that the possible reasoning for the results seen in this review is that combined uphill–downhill sprinting may provide a broader and therefore more relevant load–velocity stimulus with respect to traditional sprinting. The theories above are conceptualized in Fig. 5, showing the respective exposure and demands of the sprint training methods of this review. While more studies in the future are required to fully determine the true effect, based on the results of this meta-analysis, it does seem that combined uphill–downhill sprint programmes may be superior to resisted and traditional sprint training programmes.

Regarding the optimal slope, more research is required in this area. One acute study showed that a 5.8° decline resulted in the greatest increase in sprint velocity when sprinting downhill, possibly suggesting the gradient to achieve the greatest overspeed stimuli [24]. In our review, gradients seen were 3° and 4°. To fully understand this relationship, more investigation into the optimal slope is recommended.

4.1 Limitations

In this review, only 6 combined uphill–downhill sprinting studies (ranging from years 2006 to 2022) were included, while 16 resisted sprinting studies were included (ranging from 2009 to 2022). An analysis with more experimental data increases the chance of capturing the true effect in the population, and as such the current review is limited by its fewer combined uphill–downhill sprinting data. One result of this limitation is that combined uphill–downhill sprinting did not possess data for sprint distances below 35 m; thus, early-acceleration performance changes could not be inferred for this sprint training method. Furthermore, having fewer studies for combined uphill–downhill sprinting limited the possibility of additional subgroup analyses to be done, and so the results of this review should be considered with respect to the defined population, interventions and outcome. Additionally, the eligibility criteria of this review did not

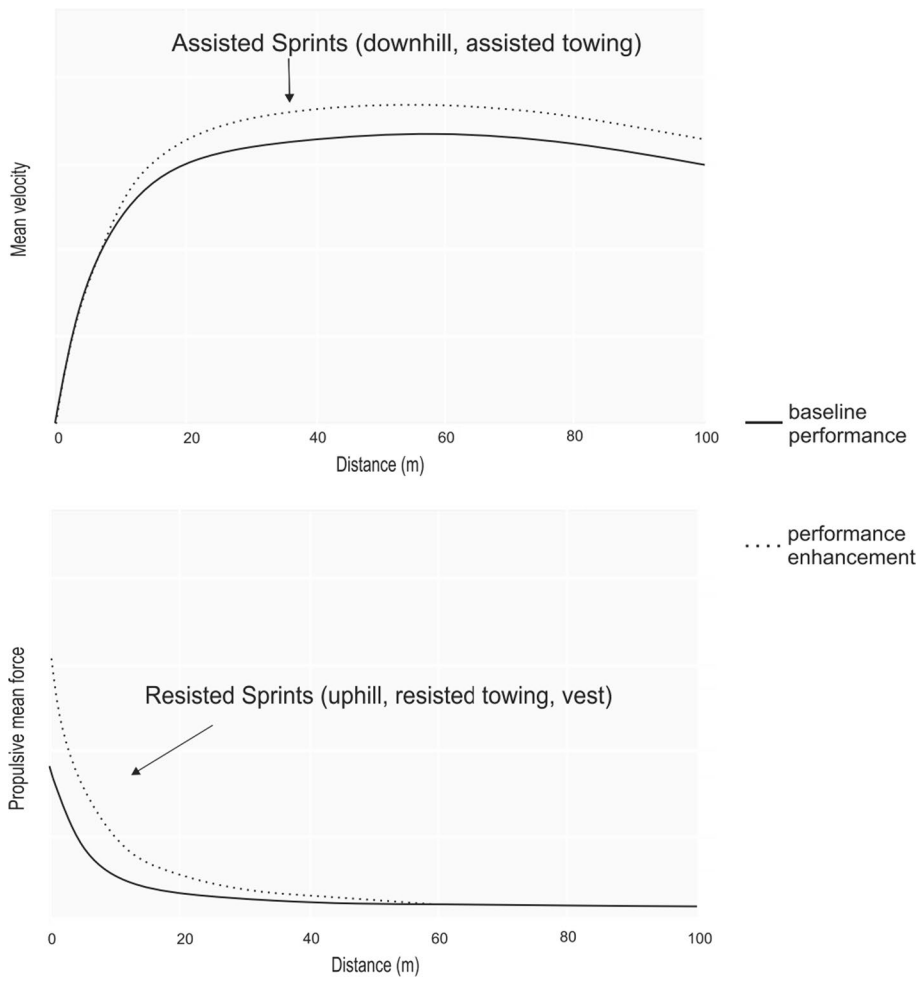

Fig. 5 Conceptualized models of an individual's baseline propulsive mean force and mean velocity capabilities during sprinting. Solid lines represent baseline performance and dashed lines represent respective enhancements from each specific sprint-training method. Resisted sprinting may expose and enhance early acceleration force production; overspeed or assisted sprinting may expose and enhance maximum velocity; it is theorized that combined methods that address both ends of the spectrum may be most appropriate and transferable to a full sprint

have restrictions in age, sex or training experience of participants; therefore, results in these groups must be considered with respect to sex, youth maturation and training level. Of note, 5 out of 6 UDS study groups contained non-practising/competing participants, while in the RS groups it was only 2 out of 17. This likely causes disparity between the observed effects as lower-level individuals can experience greater improvements in sprint performance, while magnitudes of improvements tend to taper in higher-trained individuals. This further highlights the need for further investigations into UDS with trained athletes. It is also worth noting the differences in average training programme volumes between UDS and RS. RS programmes sprinted an average of nearly 3000 m total, while UDS sprinted nearly 10,000 m. This could be in part because the UDS sprinted nearly 80 m per sprint, which led to an accumulation across sessions and weeks. Another reason could be in part due to the increased demand of towing weighted sleds which may limit how much volume can be prescribed. To remedy this, sessional training efficacy (TE_S) was included to normalize performance changes to training volume. Finally, there is a gap in longitudinal assisted sprinting studies in the literature [63] which limited the inclusion of assisted sprinting studies in this analysis. Future research in the long-term performance adaptations to assisted sprinting can provide data for a full comparison between all secondary sprint methods.

Despite this, it is important to be weary of differences in outcomes within this review considering the outlined methodological differences. Future research should directly control and compare RS against UDS to elucidate possible differences in performance adaptations.

5 Conclusion

Combined uphill–downhill sprinting and resisted sprinting methods both significantly improve sprint performance in healthy individual populations. Combined uphill–downhill sprinting showed significantly greater improvements in sprint performance compared with their traditional sprinting controls, while resisted sprinting showed no difference when compared with traditional sprinting. Both combined uphill–downhill and resisted sprinting modalities appear to be effective, and consideration of the individual's physical and technical needs is important to consider when incorporating these methods.

The findings of the present review are practically significant as combined uphill–downhill sprinting seems to provide practitioners with a specific training alternative that may be superior to resisted sprinting. When compared with resisted sprinting, combined uphill–downhill sprinting can provide both overspeed and overload options which possibly highlights its broader load–velocity stimulus. While resisted sprint methods may serve as a teaching tool to help with technique and posture, in the context of this paper, resisted sprinting performance enhancements do not seem superior to those of traditional sprinting, but more investigations are warranted. When considering these findings and the inherent logistical characteristics of each method, combined uphill–downhill sprinting may provide an advantage when resources such as budget, equipment and space are limited. The possibility of using a single hill to provide combined resisted and assisted-sprinting options to large groups of individuals, despite limited resources, reveals great value to the strength and conditioning coach and sports clubs.

Future research should directly compare all secondary methods (assisted, resisted and combined) against traditional sprint controls. Additionally, investigations into the optimal hill gradient as well as the performance changes from longitudinal assisted-sprinting interventions are also recommended by the authors.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40279-023-01916-y.

Data availability Upon a reasonable request to the corresponding author, access to the data that underlie the findings of this review can be provided.

Declarations

Funding Not applicable.

Conflict interests Maziar J. Hamad, Pedro E. Alcaraz and Eduardo Sáez de Villarreal have no competing interests relevant to the contents of this review.

Availability of data Supplementary data are included in the Electronic Supplementary Material (ESM); all other data are available upon request.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Code availability Not applicable.

Author contributions MJH and ESV synthesized the concept of the work. MJH conducted the writing, data acquisition and analysis. MJH, PA and ESV interpreted the data. MJH drafted the work, and PA and ESV revised it. All authors agreed to be accountable for all aspects of the work. All authors read and approved the final version.

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